Ali Habil

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Education

University of South Florida Tampa, FL

*B.S. in **Electrical Engineering** | **GPA: 3.92/4.0*** *Aug. 2024 – Expected Dec. 2027*

- Relevant Coursework: Semiconductor Devices, Digital Circuits, Linear Control Systems, Electromagnetics, Embedded Systems, Electronics, Signals and Systems, Electromechanical Systems, Verilog

Technical Skills

Digital & Embedded: **Verilog/SystemVerilog**, **Vivado**, **Artix-7 FPGA**, **C/C++**, ESP32, Arduino, I2C
Circuit Design & Test: **KiCad**, EasyEDA, **LTspice**, PSpice, PCB layout, **oscilloscope**, multimeter, logic analyzer, **hardware bring-up**, fault isolation
Controls & Programming: **PLC/HMI**, **MATLAB**, **Python**, Fusion 360, Git/GitHub, Linux, JavaScript, soldering

Experience

Undergraduate Research Assistant Tampa, FL

RARE Lab, University of South Florida *May 2026 – Present*

- Supported **BLOOM robotic arm** prototyping and assembly by designing and iterating **Fusion 360** components for mechanical integration of **STS3215 servos** and actuated joints.
- Identified robot integration issues through **hardware bring-up**, sensor/actuator checks, electrical connection testing, and **component-level fault isolation** to support functional prototype testing.
- Created **GitHub assembly documentation** for BLOOM Arm V2, recording hardware configurations, fastener specifications, and test observations to support repeatable builds and team handoffs.

Frontend Developer Intern Delaware / Hybrid

DataDripper *Jul. 2024 – Dec. 2024*

- Reduced reported software bugs by **40%** through **debugging**, **validation testing**, and code reviews of real-time dashboards built with **JavaScript**, **Python**, **Plotly**, and Chart.js.
- Delivered interactive **data visualization dashboards** and on-schedule updates by collaborating with cross-functional teams to reproduce defects, isolate root causes, and implement fixes.

Projects

8-Bit FPGA Accumulating Calculator | Verilog, Artix-7, Vivado Oct. 2026

- Implemented signed 8-bit accumulation by designing a **Verilog** ripple-carry adder/subtractor, **two's-complement arithmetic**, and an enable-controlled D flip-flop accumulator.
- Validated sequential addition/subtraction and positive/negative test cases using **Vivado RTL simulation**, **testbenches**, and waveform analysis; synchronized button inputs for reliable control.

Digital Overcurrent Relay with RMS Sensing and Event Logging Jun. 2026 – Present

- Built an **ESP32** protection prototype that detects overcurrent conditions using **RMS current measurement** and executes **trip logic**, relay actuation, and event logging.
- Tested sensor-to-relay operation through **hardware bring-up** and fault testing; developing a custom **KiCad PCB** layout for the next hardware iteration.

PLC Water Tank Control and HMI System 2026

- Automated tank filling and overflow protection by developing **PLC control logic** for level monitoring and filling control.
- Enabled operator monitoring by building an **HMI** that displays tank levels, filling rates, and control-system status.

SPECTER-AI Wearable Cybersecurity Alert System | Hardware Lead Apr. 2026

- Integrated a **Particle Argon**, **I2C LCD**, buttons, LEDs, buzzer, and analog input; wrote **firmware** supporting local alerts and cloud communication.
- Earned **1st Place, Tech for Good** at HackaBull 2026 after leading **hardware/firmware integration**, interface testing, and debugging.

Leadership and Involvement

Vice Chair of Projects Tampa, FL

IEEE Power & Energy Society, University of South Florida *Oct. 2026 – Present*

- Translate **smart-grid** switching and **power protection** concepts into an interactive educational game for K–8 students.

Resident Assistant Tampa, FL

USF Housing & Residential Education *Aug. 2025 – Present*

- Support a residential community of **400+ students** through **crisis response**, conflict resolution, incident documentation, and coordination with university staff.